Dependency Injection

DI · Wire Greeter without hard `new` in controllers · Build with Waqar

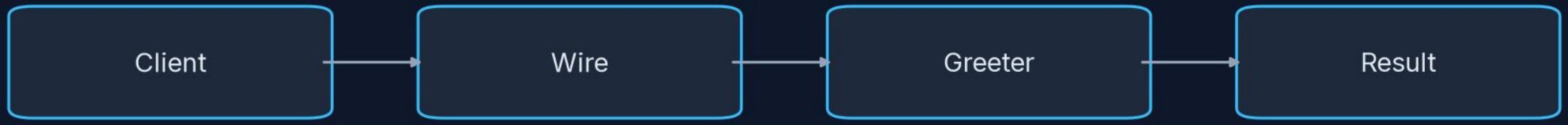
Today's Hook

- Today's force: Wire Greeter without hard `new` in controllers
- We earn DI by feeling the naive pain first.
- Demo main prints a real JDK 25 trace (happy + failure).
- Key types: Greeter, FormalGreeter, DiContainer, HelloController

The Problem

- Scenario: Wire Greeter without hard `new` in controllers
- Constraints: solve «Wire Greeter without hard `new` in controllers» with DI — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Wire Greeter without hard `new` in controllers
- Success looks like: hi Ada

Structure



DI: Wire Greeter without hard `new` in controllers

Real-World Scenario

- Running example: Wire Greeter without hard `new` in controllers
- Key types: Greeter, FormalGreeter, DiContainer, HelloController
- Happy-path stdout cue: hi Ada
- Failure-path stdout cue: caught=missing Greeter

Naive Code

```
package com.buildwithwaqar.patterns.enterprise.di.problem;

import java.util.*;

public final class NaiveController {
  public String hello(String name){ return new GreeterHard().greet(name); }
}
final class GreeterHard { String greet(String n){ return "hi "+n; } }
```

Why Naive Hurts

- Naive version fails to cleanly solve: Wire Greeter without hard `new` in controllers
- Without DI, call sites absorb the wrong details.
- Failure mode to show on camera: caught=missing Greeter
- New requirements force shotgun edits across the naive code.

Solution

```
package com.buildwithwaqar.patterns.enterprise.di.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.concurrent.atomic.*;
import java.util.concurrent.locks.*;
import java.util.function.*;

interface Greeter { String greet(String name); }
final class FormalGreeter implements Greeter { public String greet(String n){ return "Hello, "+n; } }
final class DiContainer {
  private final Map<Class<?>, Object> beans=new HashMap<>();
  <T> void register(Class<T> type, T impl){ beans.put(type, impl); }
  @SuppressWarnings("unchecked") <T> T get(Class<T> type){
    Object o=beans.get(type); if(o==null) throw new IllegalStateException("missing "+type.getSimpleName()); return (T)o;
  }
}
final class HelloController {
  private final Greeter greeter; HelloController(Greeter g){ greeter=g; }
  String hello(String name){ return greeter.greet(name); }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.enterprise.di.problem;
import java.util.*; import java.util.concurrent.*;
public final class ProblemDemo {
  public static void main(String[] args) throws Exception {
    System.out.println("=== DI: wire Greeter without hard new ===");
    System.out.println("--- naive ---"); System.out.println(new NaiveController().hello("Ada"));
System.out.println("--- DI ---");
var c=new DiContainer(); c.register(Greeter.class, new FormalGreeter());
var ctl=new HelloController(c.get(Greeter.class));
System.out.println(ctl.hello("Ada"));
try { new DiContainer().get(Greeter.class); } catch (IllegalStateException e){ System.out.println("caught="+e.getMessage()); }
  }
}
```

Walkthrough Output

```
=== DI: wire Greeter without hard new ===
--- naive ---
hi Ada
--- DI ---
Hello, Ada
caught=missing Greeter
```

Step-by-Step Trace

- Match stdout to stages — look for: hi Ada
- Then the injected failure: caught=missing Greeter
- Re-run: java -cp ... 27-dependency-injection ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```
// Java 25 direction for DI
// Prefer records + sealed where they shrink the demo.
// Problem under test: Wire Greeter without hard `new` in controllers
```

Trade-offs

- Use DI when «Wire Greeter without hard `new` in controllers» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind DI in product language.

Related Patterns

- Name the pattern usually confused with DI and separate the forces.
- Sketch the roles for «Wire Greeter without hard `new` in controllers» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Wire Greeter without hard `new` in controllers
- Naive pain → DI boundary → Demo stdout.
- Remember the failure cue: caught=missing Greeter
- Break a ProblemTest assertion, watch it fail, restore it.